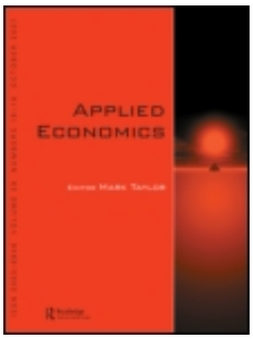

Applied Economics

ISSN: 0003-6846 (Print) 1466-4283 (Online) Journal homepage: www.tandfonline.com/journals/raec20

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To cite this article: George S. Ford (2011) An investigation into the relationship of retail gas prices on oil company profitability, Applied Economics, 43:27, 4033-4041, DOI: 10.1080/00036841003781486

To link to this article: https://doi.org/10.1080/00036841003781486

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Applied Economics, 2011, **43**, 4033–4041

Routledge
Taylor & Francis Group

An investigation into the relationship of retail gas prices on oil company profitability†

George S. Ford

The Phoenix Center for Advanced Legal & Economic Public Policy Studies, 5335 Wisconsin Avenue NW – Ste. 440, Washington, DC 20015, USA
E-mail: george.ford@phoenix-center.org

In this article, we present several empirical procedures using publicly available data from the US Department of Energy (DOE), and the Securities and Exchange Commission (SEC) to examine the relationship of oil company gross profit margins to retail gas prices. This descriptive analysis indicates that the profit margins of the major integrated oil companies are lower, on average, during periods of extremely high gas and oil prices (and, in fact, are even lower than in times of extremely low gas and oil prices). Large oil companies are most profitable during periods of moderate gasoline prices. Smaller, vertically integrated oil companies and firms, primarily in the business of refining purchased crude oil, exhibit a consistently inverse relationship between profit margins and retail gas prices – as gas prices increase, these firms become less profitable. We find no evidence for the increase in the gross profit margins of oil companies during episodes of very high retail gasoline prices.

I. Introduction

In the early half of the twenty-first century, gas prices rose well above the historical levels as crude oil prices skyrocketed. Even today, there are few signs of relief and no shortage of speculation about the cause.[1] While there are many explanations for high oil and gas prices, in political circles the blame inevitably goes to oil company greed. As the Department of Energy (DOE) study observes, '…when substantial price changes occur, especially upward, there are often allegations of impropriety, even price gouging, on the part of petroleum refiners and/or marketers (Burdette and Zyren, 2003, p. 1)'. When Hurricane Katrina disrupted refining supply in 2005, charges of price gouging escalated in the US, and political pressure against the industry and calls for profit taxes intensified. House Minority Leader, Nancy Pelosi, in April 2006 (after an increase in gas prices of about 30% that year), called oil company profits 'obscene', 'an insult to the consumer' and 'a raw deal for the American people'.[2] Senator Hillary Clinton (D-NY)

†The views expressed in this article are the author's alone and do not represent the views of the Phoenix Center, its Adjunct Fellows, or any of its individual Editorial Advisory Board members. No financial support from the oil industry for this research.

[1] A survey by *The Economist* magazine asserts that prices have risen due to 'an unprecedented combination of tight supply, surging demand, and financial speculation'. *Oil in Troubled Waters*, *The Economist* (28 April 2005).

[2] Pelosi: Republican So-Called Lobby Reform is a Ruse (27 April 2006). Available at www.releases.usnewswire.com/GetRelease.asp?id=64760.

Applied Economics ISSN 0003–6846 print/ISSN 1466–4283 online © 2011 Taylor & Francis
http://www.informaworld.com
DOI: 10.1080/00036841003781486

in May 2006 proposed new taxes on oil firms based on their profits.[3] One poll indicated at the time that 69% of Americans would favour price controls on gasoline.[4]

Political attention on the oil industry has resulted in a number of government studies on pricing and profits in the oil industry, including efforts by the DOE (2003), the Federal Trade Commission (FTC; 2005, 2007) and the Government Accountability Office (GAO; 2004, 2005, 2007), as well as some studies by individual states. While the methods and conclusions vary to some degree, for the most part these efforts 'have revealed market factors to be the primary drivers of both price increases and price spikes (FTC, 2005, p. ii)'. No study claims to provide evidence of price gouging by oil companies.

The purpose of this article is to contribute to the debate over gas prices and industry profits by deciphering the association between oil company accounting profitability and retail gas prices using publicly available data. The findings are interesting. While the general perception is that high gas prices will increase oil company profitability, the empirical analysis reveals that the gross profit margins of the major, integrated oil companies are actually *lower* during periods of extremely high gas and oil prices. In fact, the margins of these firms are lower during periods of extremely high gas prices than they are during episodes of very low gasoline prices. Profitability is highest during periods of moderate gasoline prices. In contrast, firms primarily in the business of refining purchased crude oil (along with small vertically integrated oil companies) exhibit a consistently inverse (though weak) relationship between retail gas prices and profit margins – i.e. as gas prices increase, these firms have lower gross profit margins. Setting political rhetoric aside, we find no empirical evidence that oil companies of any size or business plan systematically have higher profit margins during episodes of very high retail gasoline prices.

II. Empirical Analysis of Oil Industry Profitability

There is no doubt that the major integrated oil companies are, generally, profitable firms. It is common to see the larger oil players, such as Exxon–Mobil and Chevron, post several billion dollars of gross profits and net income. A *Washington Post* news story reported on Exxon–Mobil's First Quarter, 2006 (1Q06), profits (net income) of $8.4 billion (Mufson and Murray, 2006). While an impressively large number, this billion-dollar profit loses steam somewhat when we realize it represented only 9.7% of the $86.4 billion in revenue the company reported that quarter. If a firm has many billions in revenues, then it seems sensible that it may report billions in profits (e.g. net income) even without market power or the abuse of it.

In an effort to better understand the relationship between profits and prices, we perform some empirical analysis on publicly available data on oil company profit margins and retail gas prices in the US. The primary sample covers 45 quarters beginning in 1995 for 16 firms. The analysis is descriptive and not motivated by a particular theoretical framework, and is thus limited in some respects. The findings are, however, interesting and somewhat unexpected. We hope our results are helpful for future research and future public policy debate regarding the oil industry.

Data sources

Our database of oil company profits are derived from the financial statements filed with the Securities and Exchange Commission (SEC) including Forms 10-K (Annual Reports) and 10-Q (Quarterly Reports).[5] Our dataset consists of data from the first quarter of 1995 to the first quarter of 2006. Thus, we have about 11 years of data covering 45 quarters. Retail price data on gasoline is provided by the Energy Information Administration (EIA) of the US DOE, as is data on crude oil prices.[6] The retail price of gasoline (as reported by the US DOE) is assumed to be common to all firms. All prices are converted to

[3] Remarks of Senator Hillary Rodham Clinton at the National Press Club on Energy Policy (23 May 2006). Available at: www.senate.gov/~clinton/news/statements/details.cfm?id=255982.
[4] Much Ado About Nothing, *The Economist* (1 June 2006).
[5] Financial forms are available on the Internet, at http://www.10kwizard.com. Fourth quarter profits are necessarily calculated as the difference in annual figures less the reported financials for the 9 months ending 30 September. In a few instances, this calculation rendered results departing substantially from preceding quarters. Preliminary analysis indicates that these few peculiarities do not affect our findings.
[6] We use the composite gas price. This data and much more is available on the Internet, at http://www.eia.doe.gov/. This data is weekly, so we compute an average over the quarter to produce a quarterly retail price estimate.

constant dollars using the Consumer Price Index (CPI) (with December 2005 as the base year).[7]

An alternative approach would be to use crude oil prices rather than retail gas prices, but gas prices are chosen for a number of reasons. First, it is high gas prices that bring scrutiny to the industry. Second, the empirical results and summary statistics are more readily understood when expressed in terms of gas prices, which most readers pay. Third, it makes little difference as the price of gasoline is almost perfectly correlated with the price of West Texas Intermediate crude oil ($\rho = 0.975$). Analysis not reported here indicated that the results are not materially altered if oil rather than gas prices are used.[8]

The sample is terminated in early 2006, although data for later periods is now available. In the years subsequent to the chosen period, oil markets have been marked by what many describe as a speculative bubble (Chazan and King, 2008; Hamilton, 2009). Indeed, in January 2007 the price of crude was about \$50 a barrel, but rose to over \$130 barrel by July 2008.[9] Then, by December 2008, the price per barrel had plummeted to about \$30 as a significant economic recession hit the US and much of the rest of the world (National Bureau of Economic Research (NBER), 2008). A test of equal variance across the 1995–2006 period and the 2006–2008 period is easily rejected (F-statistic $= 3.97$, $p < 0.001$), with the variance in the post-sample period being about four-times larger than in the primary sample.[10] While some may protest the use of terms like 'speculative bubble', it is clear that the last few years may be plausibly described as an outlier in the recent history of oil markets. Nevertheless, data is collected through 2008 on the larger oil companies to evaluate the performance of the empirical tests in more recent periods. We do not claim, however, that the empirical models in this article are adequate to evaluate the very recent, highly volatile period in oil markets or measure the impact on firms of a global recession. In fact, we cannot claim the model transfers to any period outside the sample dates.

Profit measure

Profits are measured using gross profit margins. Gross margins are computed as sales minus operating expenses divided by sales, a frequently used format for the study of profitability in the economics literature (Cowling and Waterson, 1976; Long, 1982; Long and Ravenscraft, 1984; Scherer and Ross, 1990, p. 418; Bhattacharya and Bloch, 1997; Sakakibara and Yamawak, 2008) and in government studies of the oil industry (FTC, 2005, 2007; GAO, 2004). Gross margins are the most sensitive to departures of revenues from oil input costs, and are not affected by noncash expenses and accounting practices such as asset write-offs, onetime merger costs or other 'extraordinary' expenses (Long and Ravenscraft, 1984, p. 495; Scherer and Ross, 1990, p. 418; Nevo, 2001).[11] The FTC's (2007) analysis of 'price gouging' relied heavily on 'gross profit margins'.

Using financial reports, we attempt to measure costs in a similar fashion for all firms, even though firms differ in their reporting of expenses. We do not extract 'excise taxes' from revenues, since differences in reporting do not allow a consistent extraction calculation. Thus, gross margins are overstated in our analysis, but consistently. Of course, all our findings are limited to our definition of profits or profitability, and limited to the relationship between our chosen measure of profitability and retail prices.

Classifying firms

We break the oil industry down into three segments. 'Big Oil' companies include Chevron–Texaco, Exxon–Mobil, Conoco–Phillips and Shell in both their past and current incarnations (Chevron, Texaco, Exxon Phillips and Mobil).[12] The Big Oil dataset is not a balanced panel, meaning that the full set of data is not available for all firms in all quarters (since former incarnations of firms disappear as a consequence of mergers). For the full dataset of these Big Oil companies, there are 190 observations of quarterly profits and retail prices, which is an adequate sample size to represent the industry and for statistical testing. Our group of 'Small Oil' companies

[7] The CPI data is available on the internet, at http://research.stlouisfed.org/fred2/series/CPIAUCNS/9. This data is monthly, so we use the second month of each quarter as the quarterly CPI.
[8] All unreported results and details are available from the author upon request.
[9] Data from weekly spot prices in the US from http://www.eia.doe.gov/.
[10] Data from weekly spot prices in the US from http://www.eia.doe.gov/.
[11] For example, in 2008, Conoco–Phillips reported a net income of −\$16 998 due to a markdown of Goodwill of \$25 443 and an acquisition worth \$7 410. In that same year, the company reported \$65 billion in gross profits with a gross margin of 27%, which was very similar to its 2007 results where it reported a net income of \$11 891 in the absence of the one-time accounting adjustments.
[12] We exclude Conoco (prior to Conoco–Phillips merger) because of irreconcilable inconsistencies in the reported figures in their 10-K and 10-Q forms. Conoco–Phillips data begins in year 2003 following the merger of Conoco and Phillips.

includes Marathon, Hess, Sunoco, Giant Frontier, Tesoro and Valero. The Small Oil companies are further divided into two minor segments. Marathon and Hess are vertically integrated firms involved in exploration, refining and retail, and we describe these as 'Vertically Integrated' firms. The remaining firms, including Sunoco, Giant, Frontier, Tesoro and Valero, are primarily refineries with some retail distribution, so we will refer to these as 'Refiners' throughout the rest of this article.

The distinction between vertically integrated firms, such as the Big Oil and Vertically Integrated firms, is an important one for theoretical reasons. For refineries and retail firms, an increase in the price of oil is simply an increase in the price of an input. As such, we would expect, if anything, a negative relationship between profit margins and gas prices. For the vertically integrated firm, oil is both an output and input as long as some of its refined oil is purchased, rather than self-produced. As oil prices rise, therefore, profits may rise for one segment of the firm (oil extraction) but fall for another (oil refining and retail sales). A nonlinear relationship may be observed as the firm may switch between 'mostly seller' to 'mostly buyer' of crude oil. Observing a nonlinear relationship between prices and profits for vertically integrated firms, both big and small, lends support to this hypothesis.

III. Analysis of 'Big Oil' Profits

To provide a general feel for the relationship between margins and gas prices, we begin by comparing the average gross profit margins at each price. These margins, illustrated in Fig. 1, are filtered for seasonal and fixed effects, and averaged across firms.[13] On the horizontal axis is the inflation-adjusted retail price for gasoline (covering its full range of \$1.19–\$2.63); gross margins are on the vertical axis. At first glance, we see a nonlinear relationship between margins and prices, and this relationship is confirmed in additional analysis.

Next, we compute the average margins for periods of 'moderate', 'high' and 'low' prices. We define

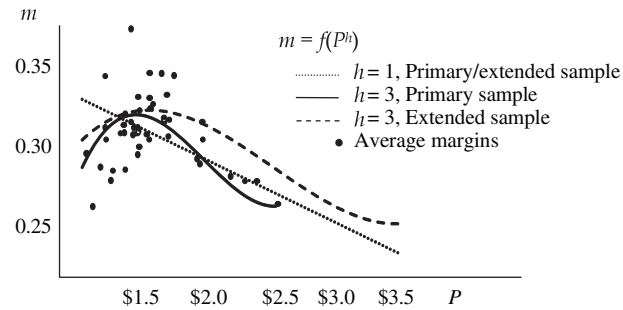

Fig. 1. Average profit margins and retail price ('Big Oil')

'high' and 'low' prices as either the highest and lowest 25% of real prices or the highest and lowest 10% of prices. There are five observations, each categorized as the highest or lowest 10% of prices, 11 observations of the highest 25% of prices (29%) and 11 observations of the lowest 25% of prices.

Table 1 summarizes the average profit–sales ratios for the sample under the different definitions of prices.[14] We define periods of low and high prices by reference to the average price of gasoline (as reported by the DOE) from 1995 to the first quarter of 2006. Gasoline prices during this decade averaged \$1.68 per gallon, with a minimum and maximum value of \$1.20 and \$2.63 and an SD of 0.32. In quarters with the lowest 10% of prices, the average retail price was \$1.28 per gallon, and the mean was \$2.37 for the highest 10% of prices. The average real price during times of moderate gasoline prices during this decade was \$1.62.

From Table 1, we see that, on average, the gross profit margin for the Big Oil group during moderate prices is 32.5%.[15] The table also reveals that profit margins are lower during periods of extremely low and high retail gas prices, as is also shown in Fig. 1. During periods where prices are in the bottom 25% of prices over the sample period, the profit margin falls slightly to 32.3%, but the reduction is not statistically different from zero at standard levels. For the lowest 10% of prices, the average profit margin is 30.5%, which is 5% lower than the profit margin during periods of moderate prices. This difference is statistically significant at better than the 5% level ($t = -2.85$). Profit margins also fall when prices are

[13] This is accomplished by regressing the margins on seasonal and firm-specific dummy variables. The common intercept of this regression is then added to each firm's residual, then these residuals are averaged across the firms.

[14] These are simple means differences, but we use Generalized Least Squares (GLS) with fixed effects and panel-corrected SEs as a computational device. We assume cross-sectional heteroscedasticity with each cross-section having its own residual variance. The seasonality adjustment includes dummy variables for the first three quarters of the year, but we note that the general findings are unaffected by the adjustment. Dummy variables are used for different price categories. The t-statistic on the coefficient of these variables is a direct test of statistical significance.

[15] The mean of the 'moderate' average price changes slightly between the high/low 25% and 10% definitions, but not by large enough amounts after rounding to report the difference.

Table 1. Average gross profit margins in periods of high/low gas prices

Firm type	Lowest 10% prices ($P = \$1.28$)	Lowest 25% prices ($P = \$1.42$)	Moderate prices ($P = \$1.62$)	Highest 25% prices ($P = \$1.93$)	Highest 10% prices ($P = \$2.37$)
Big Oil	0.305	0.323	0.325	0.308	0.264
Small Oil	0.143	0.163	0.141	0.140	0.120
Vertically Integrated	0.273	0.330	0.286	0.272	0.226
Refiners	0.095	0.105	0.086	0.092	0.079

very high. For the top 25% of prices over the sample period, the average margin is 30.8%, a 6% reduction from the profit margin at moderate prices. At the 10% highest prices, the average profit margin is 28.2%, or 13% below the margin during periods of moderate prices. Both these margin reductions are statistically different at better than the 5% level ($t = -3.08, -3.07$). Observe from Table 1 that the difference in mean profit margins increases as the prices get more extreme. This result is encouraging, since it is consistent with the monotonicity of the underlying relationship between prices and profits. Note also that the effect of retail price on profit margins is asymmetric in that very high prices reduce profits more than very low prices (about 15% lower at high prices versus 5% at low prices).

This initial evaluation, both Fig. 1 and Table 1, suggests a nonlinear correlation and asymmetric association between prices and profits for these large, vertically integrated oil firms. Both very low and very high prices are bad for profitability of these Big Oil firms. The highest profit margins occur when gas prices are in the intermediate retail price range. Nevertheless, to draw firmer conclusions, the results suggested by Table 1 need to be tested further, by means of a regression analysis allowing for a continuous relationship between margins and prices and a more formal test of nonlinearity.

Regression model of 'Big Oil' profits

In this section, we estimate the continuous relationship between profitability and retail prices for our Big Oil firms by means of a regression analysis. To estimate this relationship, we can use the following equation:

$$m_{i,t} = \sum_{h=1}^{H} \beta_h P^h + \sum_{i=1}^{m} \gamma_i FIRM + \sum_{j=1}^{g} \delta_j QTR + \varepsilon_{i,t} \quad (1)$$

where $m_{i,t}$ is the gross margin of firm i in period t, where P^h is retail price in period t raised to power h, $FIRM$ is an intercept for each of the m firms in the sample (fixed effects estimation), QTR is g dummy variables for each quarter (three included to avoid the dummy trap) and ε is an econometric disturbance term. We can allow for nonlinearity by including higher powers (up to H) of the variable P^h. The coefficients β_h are H coefficients on the price variables that can be used to assess the relationship between profit margins and retail prices. Using a variety of panel unit root tests, the null of a unit root for the profit margins is rejected (Breitung, 2000; Levin *et al.*, 2002; Im *et al.*, 2003). An evaluation of the Durbin–Watson statistics did not reveal the presence of serial correlation in the data.

Using the Big Oil sample, we first consider four functional forms to evaluate the robustness of the estimated relationship: lin–lin, log–lin, lin–log and log–log form. The estimation method is fixed effects with the assumption of cross-section heteroscedasticity (i.e. feasible GLS), and robust panel-corrected SEs are employed for statistical testing. Table 2 summarizes the regression results of Equation 1 where a linear relationship between profits and price is assumed (so $H = 1$). While our initial analysis (shown in Table 1) indicates that the relationship between gas prices and profitability for these firms is probably nonlinear, Table 2 presents the results from this estimation assuming linearity.

Table 2 shows that gross profit margins for our Big Oil companies and retail gas prices are negatively related (on average) over the period 1995 to first-quarter 2006. (This negative and statistically significant relationship is robust to functional in form, including lin–log, log–log and log–lin.) The margin-price elasticity is -0.20, meaning a 2% reduction in margins is expected from a 10% increase in real prices. The coefficients on the quarterly dummy variables indicate that the fourth quarter typically has the lowest margins, though only one of the quarterly dummies is statistically significant. From the fixed effects estimates, we see that Exxon and Exxon–Mobil are generally much more profitable than the other oil companies in the sample are, and Texaco is significantly less profitable than its peers are.

Next, we allow for a nonlinear relationship between prices and profits, as inferred by the exploratory analysis above. Nonlinearity in the relationship

Table 2. Regression results, nonlinear specifications ('Big Oil')

Variable	Linear[d] Coefficient (*t*-statistic)	Third-degree polynomial[c,d] Coefficient (*t*-statistic)
P	−0.0368	1.253
	(−5.99)[a]	(4.68)[a]
P^2	–	−0.637
		(−4.29)[a]
P^3	–	0.102
		(3.83)[a]
$QTR1$	0.005	0.009
	(1.05)	(2.76)[a]
$QTR2$	0.009	0.008
	(1.89)[b]	(2.23)[a]
$QTR3$	0.008	0.008
	(1.64)	(2.34)[a]
Fixed effects		
Common intercept	0.375[a]	−0.469[a]
Chevron–Texaco	−0.002	−0.008
Chevron	−0.031	−0.034
Texaco	−0.174	−0.177
Exxon–Mobil	0.138	0.122
Exxon	0.160	0.159
Mobil	0.067	0.065
Shell	−0.046	−0.046
Conoco–Phillips	−0.034	−0.032
Phillips	−0.049	−0.053
Unweighted R^2	0.86	0.88

Notes: [a]Statistically significant at the 5% level or better.
[b]Statistically significant at the 10% level or better.
[c]Statistical testing indicates that this is the preferred model.
[d]Dependent variable = m.

is introduced by increasing H in Equation 1 to a value larger than one. We select H by statistical procedure, increasing it to the point where a variable-addition F-test indicates that the addition of higher order P terms does not increase the explanatory power of the regression (Gujarati, 1995: 253–8). This procedure indicates the best value of H is 3 (a third-degree polynomial).

As shown in Table 2, the estimated regression confirms a nonlinear relationship as suggested by Fig. 1 and Table 1. In short, profits are lower when gas prices are either extremely low or high. The profitability of these Big Oil firms is at its peak when gas prices are slightly below the sample mean price (the polynomial is maximized at a price of \$1.59, whereas the sample mean is \$1.68).

Figure 1 illustrates the estimated relationship between profits and prices. The shape of the curve for the third-degree polynomials is consistent with the plotted margins in Fig. 1 and means in Table 1. The asymmetric shape of the curve parallels the evidence, suggesting that a negative relationship between prices and profits is more pronounced at high, rather than low, prices. The nonlinear relationship found in the data suggests that costs are the more significant driver of very high prices, since gross margins are lower when prices are very high. These results are inconsistent with the political notion that high prices are evidence of price gouging by oil companies.

IV. An Analysis of 'Small Oil' Companies

Firms in the Big Oil sample are all vertically integrated into oil production, refining and retailing and are large firms by any financial measure. However, the oil industry contains a host of other firms that are not as large and which might focus on particular segments of the industry, such as refining and retail gasoline distribution. Therefore, we have also constructed a sample of Small Oil companies, which consists of seven firms: Frontier, Marathon, Sunoco, Giant, Hess, Tesoro and Valero. Within this group of Small Oil companies, some are vertically integrated and others are primarily in the business of refining and selling petroleum products. As a result, for further analysis, we break these seven companies into two groups: Vertically Integrated firms (Marathon and Hess) and Refiners (Frontier, Sunoco, Giant, Tesoro and Valero).

Table 1 includes a means-difference test for these firms identical to that for Big Oil companies. Like our Big Oil sample, the profit margins for the Small Oil sample fall when retail prices are high, falling from a profit margin of 14.1% at moderate prices to 12% at the largest 10% of prices (the difference is statistically significant at the 5% level). The same nonlinear shape observed for Big Oil appears for this group of firms, though the highest margins occur at lower prices than for Big Oil. At the lowest 25% of prices, the margin rises to 16.3% (statistically different at the 5% level). However, at the lowest 10% of prices, the profit margin is 14.3%, which is essentially equal to the margin at moderate prices. The rise in margins for the lowest 25% of prices, but not for the lowest 10% of prices, suggests a nonlinear relationship similar to that for the Big Oil sample.

Splitting the Small Oil sample into the vertically integrated and refining components provides for some interesting findings. For the two vertically integrated firms, we find a nonlinear relationship between profit margins and gas prices similar to that of the large, vertically integrated firms like Exxon-Mobil. For these smaller firms, it appears that the largest margins are earned when prices are in the low

to moderate range (33.0%, 28.6%, which are statistically unequal at the 10% level only), but margins fall when prices are very low (27.3% at the lowest 10% of prices, again statistically significant at the 10% level).[16] Table 1 clearly demonstrates that for our sample of smaller, vertically integrated firms, profitability declines as retail gas prices rise, with margins falling from 28.6% to 27.2% at the top 25% of prices and 22.6% at the top 10% of prices (the latter difference is statistically significant at the 5% level).

The refining and retail companies (Sunoco, Frontier, Giant, Tesoro and Valero) show a weak and near linear relationship between margins and retail prices: margins are highest at the lowest retail prices (about 9.5%) and fall as retail prices increase (down to 7.9% at very high prices). Relative to the other margin changes, however, the refiner profit margins are the least-correlated with retail price. In fact, none of the differences in the mean profit margins relative to the moderate price margin is statistically different from zero at the 5% level (the slight rise at the lowest 25% of prices is significant at the 10% level).

As we did with the Big Oil sample, we can estimate continuous relationships between margins and retail prices with the choice of H from Equation 1 being based on statistical testing.[17] While the means in Table 1 suggest a similar nonlinearity for the small firms as for Big Oil, the F-tests for adding either or both the squared and cubed terms of the price variable indicate that the linear model is the best specification for the Small Oil sample and its components. Statistically, we cannot reject a consistently negative relationship between profit margins and retail gas prices for these firms.

Table 3 summarizes the regression results ($H = 1$) for our 'Small Oil' groups and shows a negative and statistically-significant relationship between profitability and retail prices for the complete Small Oil sample and the Vertically Integrated sample. For Refiners, retail prices and profit margins are negatively related, but the relationship is so weak as we cannot reject the hypothesis that the relationship is zero. The margin-price elasticities (computed at the mean price) for the three sample groupings are: (a) the Small Oil sample, −0.22 (about the same as for the Big Oil group); (b) the Vertically Integrated sample, −0.42 (about twice as large as the Big Oil group) and (c) the Refiner sample, −0.08 (which is not statistically different from zero).

Table 3. Regression results ('Small Oil' sample)

	Small Oil	Vertically integrated	Refiners
Variable	Coefficient (t-statistic)[b]	Coefficient (t-statistic)[b]	Coefficient (t-statistic)[b]
P	−0.0190	−0.06223	−0.00468
	(−2.22)[a]	(−2.46)[a]	(−0.51)
$QTR1$	0.004	−0.011	0.008
	(0.46)	(−0.51)	(0.99)
$QTR2$	0.017	−0.048	0.0030
	(2.14)[a]	(−2.10)[a]	(3.46)[a]
$QTR3$	0.007	−0.054	0.018
	(0.88)	(−2.39)[a]	(2.06)
Fixed effects			
Common intercept	0.172	0.384	0.096
Marathon	0.098	0.006	–
Hess	0.110	−0.006	–
Sunoco	0.046		0.091
Giant	−0.046		−0.001
Frontier	−0.074		−0.030
Tesoro	−0.079		−0.035
Valero	−0.084		−0.039
Unweighted R^2	0.90	0.94	0.92
Mean margin	0.147	0.251	0.102

Notes: Unweighted R^2 computed as the squared correlation coefficient of the actual and fitted values of the dependent variable.
[a]Statistically significant at the 5% level or better.
[b]Dependent variable = m.

To summarize, the statistical analysis presented here suggests that for Small Oil companies, profit margins are at their lowest when gas prices are high. A simple means difference analysis indicates that smaller firms have a nonlinear relationship between margins and prices similar to Big Oil, but regression analysis indicates that a linear relationship is statistically preferred. Refiners have profit margins that are only weakly related to changes in retail gasoline prices, the relationship is not statistically different from zero.

The pattern of profit margins for the Small Oil sample is shown to follow much the same pattern as those for the Big Oil firms. Yet, smaller oil companies are not generally the target of political rhetoric. It appears, then, that in political discourse it is the absolute size of the profits that draws attention rather than the actual profitability of selling refined oil products, such as gasoline. From the perspective of profitability, the Big Oil firms are not much different from their smaller counterparts. It is the sheer size of

[16] This sample group has a smaller sample size (90 observations) than the other groupings, so the weaker significance levels could be the result of type II error (a failure to reject a true null hypothesis).
[17] We reject the presence of a unit root for 'Small Oil' and its two components.

their absolute profits that draws political and media attention.

V. Extending the Sample for the 'Big Oil' Group

As discussed above, the sample period used for this analysis covers the period 1995 to the first quarter of 2006. Since the initial study was conducted, more data has become available. However, subsequent quarters were excluded from the analysis due to the highly volatile period for oil markets beginning in 2006 and a global recession starting in 2008. These events had a significant effect on oil and gas prices. For example, the variance of oil prices in the 2006 to 2008 period was about four times larger than the chosen sample period. It may be interesting, however, to consider the impact of including data that is more recent in the analysis, through recognizing the significant differences between the two time periods. To do so, the sample for Big Oil was extended to include data through the fourth quarter of 2008. The extended sample adds 11 quarters of data. In eight of these quarters, the gas price exceeds the maximum price from the primary sample, so this additional data is truly out of the primary sample range.

If the results from the preceding analysis apply to this more recent, high price period, we expect relatively low profit margins during these eight 'very high price' periods of data. Based on the same computational device used to generate the statistics in Table 1, the average profit margin for prices exceeding \$2.63 for the extended sample is 28.6%, which is nearly equal to the 28.2% margin at the highest 10% of prices for the primary sample. The difference between these two average margins is not statistically significant, but the 28.6% margin is statistically different from the margin at moderate prices (32.5%) at better than the 5% level. These results are consistent with the findings from the earlier period in that profit margins are lower when prices are higher. Figure 1 also shows a flattening out of margins at higher prices.

Next, we estimated the model with $H = 3$ for the extended sample to see if the nonlinear relationship is retained. Since the new data has gas prices outside the range of the primary sample, we have little expectation that the coefficients will be identical given the polynomial nature of the model. As shown in Table 4, the coefficients are different, but the signs and significance levels of the price coefficients remain the same. The model with $H = 3$ is again confirmed by statistically testing. The quadratic is maximized at a price of \$1.69, which is only a few cents more than that for the primary sample and effectively equal to the sample mean of \$1.68. Figure 1 illustrates the difference in the predicted margins between samples and confirms visually the results that are qualitatively the same – the margins are lower at very high and very low gas prices.

Table 4. Regression results, extended sample ('Big Oil')

Variable	Coefficient (t-statistic)
P	0.384
	(3.91)[a]
P^2	−0.169
	(−3.69)[a]
P^3	0.022
	(323)[a]
$QTR1$	0.009
	(2.30)[a]
$QTR2$	0.007
	(1.80)[b]
$QTR3$	0.009
	(2.28)[a]
Fixed effects	
Common intercept	−0.053
Chevron–Texaco	−0.008
Chevron	−0.034
Texaco	−0.177
Exxon–Mobil	0.122
Exxon	0.159
Mobil	0.065
Shell	−0.046
Conoco–Phillips	−0.032
Phillips	−0.053
Unweighted R^2	0.87

Notes: [a]Statistically significant at the 5% level or better.
[b]Statistically significant at the 10% level or better.
[c]Dependent variable = m.

VI. Conclusion

Attacks on the oil industry for 'obscene' or 'excess' profits rise with the price of gas. In this article, we evaluate from an empirical perspective the relationship between retail gas prices on oil company profitability. Our analysis suggests that contrary to political rhetoric and media coverage, on average, Big Oil company profitability is lower during periods of very high (and very low) retail gasoline prices. Industry profitability is highest in periods of moderate gas prices. In addition, smaller, vertically

integrated oil companies and those firms primarily in the business of refining crude purchased from others and selling it at the pump have profit margins that decline as retail gas prices rise, though the relationship is very weak for refiners. Charges that the oil industry somehow unfairly profits from high gas prices appear to be unsupported by the data.

These findings are, of course, only the starting point of a complete analysis. The purpose of this article is not to answer all questions regarding profits in the oil industry, but to evaluate the price–profit relationship in a simple yet descriptively sound manner. A complete exploration as to whether oil firms are gouging customers would need to explore more than simply accounting profits, including a detailed review of retail prices and input costs. That said, our finding that profit margins fall during periods of high retail gas prices is difficult to reconcile with collusive behaviour in the industry during periods of exceptionally high gas prices.

Acknowledgements

The author greatly appreciates the helpful comments of Mark Taylor (editor), two anonymous referees, Bill Brummett, Tom Koutsky, Larry Spiwak, and John Jackson. Remaining errors are the author's responsibility alone.

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